

Trainer Debrief — Lab 5

CTQ Trees and CTQ Flowdown to Controllable Xs

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Translate VOC into measurable CTQs and cascade a business Y down to controllable Xs (A1, A4).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-05-ctq-tree-flowdown.xlsx

The flowdown must end at Xs a shift supervisor can change before lunch: fixture clamp pressure (bar), weld dwell time (s), electrode tip age (strokes) and seal ring durometer (Shore A). Ambient temperature and humidity are NOISE — you can measure them but not economically control them, so they belong in a robustness study (Lab 25), not a control plan. A CTQ that stops at 'improve quality' has not been flowed down. Note how the burst pressure measure, target and spec here are the SAME numbers used in Labs 14, 23-25 and 27 — that is the flowdown holding together across the whole project.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Every CTQ has an operational definition, a target and specification limits, and your flowdown reaches at least one layer of Xs that are directly controllable and testable.

Where this sits in the capstone

A customer need is not measurable; a CTQ is. Re-derive the CTQ tree in full, then go beyond Green Belt with CTQ FLOWDOWN — cascading a business-level Y through sub-Ys until you reach process Xs a team can actually control. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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